



SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)
Department of Mathematics and Computer Science

CIE-II SOLUTIONS

Topics: Continuous Probability Distribution & Joint Probability Distribution

UDSC-201: Probability and Statistics	Instructor: Dr. D Bhanu Prakash	Date: 09-09-2026
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1. The total area under any probability density function, over its entire range, is always equal to 1. [1 Mark | BTL-1] **Answer:** True
2. The cumulative distribution function $F(x)$ of a continuous random variable is a decreasing function of x . [1 Mark | BTL-1] **Answer:** False — $F(x)$ is a non-decreasing (monotonically increasing) function of x .
3. If two random variables X and Y are independent, then $\text{Cov}(X, Y) = 0$. [1 Mark | BTL-2] **Answer:** True
4. The mean and variance of the standard normal distribution are 0 and 1 respectively. [1 Mark | BTL-1] **Answer:** True
5. For a continuous random variable X , $P(X = a) =$ _____. [1 Mark | BTL-1] **Answer:** 0
6. The Exponential distribution is a special case of the _____ distribution when the shape parameter $k = 1$. [1 Mark | BTL-2] **Answer:** Gamma (Erlang)
7. The marginal density function of X is obtained by _____ the joint density function $f(x, y)$ with respect to y . [1 Mark | BTL-2] **Answer:** integrating
8. The normal approximation to the Binomial distribution $B(n, p)$ is considered good when $np \geq 5$ and $n(1 - p) \geq$ _____. [1 Mark | BTL-1] **Answer:** 5
9. Match the distribution in Column A with its typical property / application in Column B. [2 Marks | BTL-2]

Column A	Column B
(a) Uniform distribution	1. Models time-to-failure / reliability of components
(b) Weibull distribution	2. Bounded on $[0, 1]$; models proportions and probabilities
(c) Lognormal distribution	3. Every value in an interval is equally likely
(d) Beta distribution	4. Log of the variable is normally distributed; arises from products of positive random variables

Answer: (a) $\rightarrow$ 3, (b) $\rightarrow$ 1, (c) $\rightarrow$ 4, (d) $\rightarrow$ 2

Q10. (a) Define a probability density function (pdf) and state its defining properties. (2) (b) Write the pdf, mean and variance of the Uniform, Normal and Gamma distributions. (3) [2+3 Marks | BTL-1]

Answer:

(a) If X is a continuous random variable, a function $f(x)$ is called its probability density function if:

- $f(x) \geq 0$ for all x

$$\int f(x) dx = 1$$

- For any $a \leq b$, the following holds:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

(b)

Distribution	pdf $f(x)$	Mean	Variance
Uniform(a, b)	$f(x) = \frac{1}{b-a}, \quad a < x < b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal(μ, σ^2)	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Gamma(k, λ)	$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)}, \quad x \geq 0$	$\frac{k}{\lambda}$	$\frac{k}{\lambda^2}$

Q11. (a) Derive the mean and variance of the Exponential distribution with parameter λ . (3) (b) Explain how the Erlang, Gamma and Exponential distributions are related. (2) [3+2 Marks | BTL-2]

Answer:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Integrating by parts ($u = x, dv = \lambda e^{-\lambda x} dx$):

$$E(X) = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

$$E(X) = [-x e^{-\lambda x}]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}$$

Similarly, by repeated integration by parts:

$$E(X^2) = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = \frac{2}{\lambda^2}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

Hence Mean = $1/\lambda$ and Variance = $1/\lambda^2$.

(b) The Erlang distribution is the Gamma distribution restricted to an integer shape parameter k ; the sum of k independent and identically distributed Exponential(λ) random variables is an Erlang(k, λ) random variable. When $k = 1$, the Erlang/Gamma distribution reduces exactly to the Exponential(λ) distribution.

Q12. Explain the concept of marginal and conditional probability distributions for two jointly distributed continuous random variables X and Y, and state the condition for their independence.

[3 Marks | BTL-2]

Answer:

Marginal distribution: The marginal pdf of X is obtained by integrating the joint pdf over all values of Y:

$$f_X(x) = \int f(x, y) dy$$

and similarly:

$$f_Y(y) = \int f(x, y) dx$$

Conditional distribution: The conditional pdf of Y given X = x is defined (for $f_X(x) > 0$) as:

$$f(y | x) = \frac{f(x, y)}{f_X(x)}$$

It describes the distribution of Y once the value of X is known, and similarly $f(x | y) = f(x, y) / f_Y(y)$.

Independence: X and Y are said to be statistically independent if and only if the joint pdf factors as the product of the marginals for all x, y:

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

Equivalently, X and Y are independent if the conditional distribution of Y given X does not depend on x, i.e. $f(y | x) = f_Y(y)$.

Q13. (a) The lifetime of an electronic component is exponentially distributed with mean life 100 hours. Find (i) $P(X > 150)$ and (ii) $P(50 < X < 150)$. (1) (b) If $X \sim N(50, 100)$, find $P(45 < X < 65)$ using standard normal tables. (1) (c) A fair coin is tossed 100 times. Using the normal approximation to the Binomial distribution, find the probability of getting more than 60 heads. (1)

[1+1+1 Marks | BTL-3]

Answer:

$$\text{Mean} = \frac{1}{\lambda} = 100 \Rightarrow \lambda = 0.01$$

(i)

$$P(X > 150) = e^{-\lambda(150)} = e^{-1.5} = 0.2231$$

(ii)

$$P(50 < X < 150) = e^{-0.5} - e^{-1.5} = 0.6065 - 0.2231 = 0.3834$$

$$(b) \mu = 50, \sigma = \sqrt{100} = 10$$

$$Z_1 = \frac{45 - 50}{10} = -0.5, \quad Z_2 = \frac{65 - 50}{10} = 1.5$$

$$P(45 < X < 65) = \Phi(1.5) - \Phi(-0.5) = 0.9332 - 0.3085 = 0.6247$$

(c)

$$n = 100, p = 0.5 \Rightarrow \mu = np = 50, \sigma^2 = np(1 - p) = 25, \sigma = 5$$

Using continuity correction:

$$P(X > 60) \approx P\left(Z > \frac{60.5 - 50}{5}\right) = P(Z > 2.1) = 1 - \Phi(2.1) = 0.0179$$

Q14. The joint pdf of two random variables X and Y is given by $f(x, y) = x + y$, $0 < x < 1$, $0 < y < 1$ (and 0 elsewhere). (a) Verify that $f(x, y)$ is a valid joint density. (1) (b) Find the marginal densities of X and Y. (1) (c) Check whether X and Y are independent. (1) (d) Find $\text{Cov}(X, Y)$ and the correlation coefficient $\rho(X, Y)$. (1) [4 Marks | BTL-4]

Answer:

$$f(x, y) = x + y, \quad 0 < x < 1, \quad 0 < y < 1$$

(a)

$$\int_0^1 \int_0^1 (x + y) dx dy = \int_0^1 \left(\frac{1}{2} + y\right) dy = \frac{1}{2} + \frac{1}{2} = 1$$

and $f(x, y) \geq 0$ on the given region. Hence $f(x, y)$ is a valid joint pdf.

(b)

$$f_X(x) = \int_0^1 (x + y) dy = x + \frac{1}{2}, \quad 0 < x < 1$$

By symmetry:

$$f_Y(y) = y + \frac{1}{2}, \quad 0 < y < 1$$

(c)

$$f_X(x) \cdot f_Y(y) = \left(x + \frac{1}{2}\right)\left(y + \frac{1}{2}\right) \neq x + y$$

Since $f(x, y) \neq f_X(x) \cdot f_Y(y)$, X and Y are NOT independent.

(d)

$$E(X) = \int_0^1 x\left(x + \frac{1}{2}\right) dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

By symmetry, $E(Y) = 7/12$.

$$E(X^2) = \int_0^1 x^2\left(x + \frac{1}{2}\right) dx = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

$$\text{Var}(X) = \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{11}{144}$$

By symmetry, $\text{Var}(Y) = 11/144$.

$$E(XY) = \int_0^1 \int_0^1 xy(x + y) dx dy = \frac{1}{3}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{3} - \left(\frac{7}{12}\right)^2 = -\frac{1}{144}$$

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{-1/144}{11/144} = -\frac{1}{11} \approx -0.0909$$

The small negative correlation indicates a weak negative linear association between X and Y.